Mark Verma

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Education

ABV-IIITM Gwalior — *Madhya Pradesh, India*
B.Tech, Computer Science — SGPA 8.67 — *Graduated 2024*
Relevant coursework: **Artificial Intelligence, Statistics, Cloud Computing**

Experience

314e Corp — *Bengaluru, India*
Data Science Engineer — *Jul 2024 – Present*

- Architected and automated a full-scale **MLOps pipeline** for a patient entity-extraction model, raising accuracy from **72% to 94%**; orchestrated fault-tolerant workflows with **Temporal** and deployed on **SkyPilot**, **Hugging Face Endpoints**, and **RunPod** with a human-in-the-loop system for continuous refinement.
- Re-engineered legacy Python scripts into scalable **PySpark** jobs for large-scale **FHIR** resources across multiple clients, cutting data-loading time by **~80%**.
- Cut model training time from **90 hours to 25 hours** on >500 GB datasets via targeted hyperparameter tuning, and eliminated OOM errors by building a custom memory-efficient iterable dataloader.
- Reduced model cold-start time by **77%** by optimizing the serving process and pre-packaging assets into the runtime.
- Built a rigorous test suite to benchmark **Vision-Language Models (VLMs)** out-of-the-box, logging results on **ClearML** for transparent, reproducible comparisons.
- Reported a critical bug in Google's **Gemma 3** VLM (Flash-Attention SDPA implementation), and authored a technical blog post on the AI-native document-processing system.

Banach Technologies — *Singapore (Remote)*
Machine Learning Intern — *Apr 2024 – Jul 2024*

- Engineered and optimized **high-frequency trading** strategies, reducing data-processing latency through targeted performance tuning.
- Built and deployed ML models from scratch against live market APIs to execute hedging strategies, backed by a comprehensive unit-test suite for core trading libraries.

Alemeno — *Maharashtra (Remote)*
Software / ML Intern — *Jul 2023 – Apr 2024*

- Built a data-preprocessing pipeline for **semantic-segmentation** models over large-scale GIS raster data, and implemented distributed computer-vision algorithms (Douglas–Peucker, Jarvis March).
- Deployed scalable **Django** apps on **AWS** with an end-to-end **CI/CD** pipeline, and hardened a high-performance raster-processing app through refactoring and unit tests.

Bewgle — *Bengaluru (Remote)*
Data Analyst — *May 2022 – Jul 2022*

- Applied **NLP** techniques to proprietary Amazon-review datasets to deliver product-trend insights and predictive models; automated preprocessing in Python/Bash, cutting data redundancy by **90%**.

Projects

rumpy — NumPy in pure Rust — *Rust* — *2025*

- Building a **zero-dependency** ndarray library in pure **Rust** (flat-buffer / shape / strides layout), reimplementing broadcasting, axis reductions, and matmul from first principles to mirror NumPy's internals.
- Showing that memory-access order dominates arithmetic by beating a naive matmul via cache-friendly **ikj** loop ordering, benchmarked against BLAS-backed NumPy; spec-driven with ~30 milestone tests over Rust ownership, traits, and **Result** error handling.

ML-from-scratch — *PyTorch* — *2025*

- Implemented core ML/DL models (backpropagation, CNNs, transformers) from first principles in pure **PyTorch**, without high-level abstractions.

Resume Parser — *PyTorch, LLMs, HuggingFace* — *2024*

- Built an end-to-end **LLM**-powered resume parser with a modular OOP codebase and a **Streamlit** UI producing structured JSON output.

Medical Waste Classification — *TensorFlow, JavaScript* — *2023*

- Trained a **CNN** reaching **98.4%** accuracy, shipped it as an interactive **TensorFlow.js** web app, and published the findings in **IEEE Xplore**.

Skills

Languages: Python, C++, Rust, Bash, JavaScript
Libraries & Frameworks: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, Rasterio, Matplotlib, Django, Flask, FastAPI, Dash, vLLM, SGLang, RunPod SDK, SkyPilot
Tools & Platforms: Docker, Kubernetes, AWS SageMaker, MLflow, ClearML, Temporal, Git, Linux, PostgreSQL, Streamlit
Certifications: DevOps on AWS Specialization • GCP Network Deployment • Open-source contributor @ PyMC